

Ethanol Blending in Gasoline - Honduras

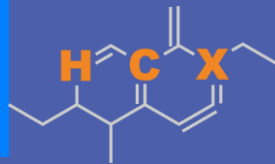
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Ethanol Blending in Latin America

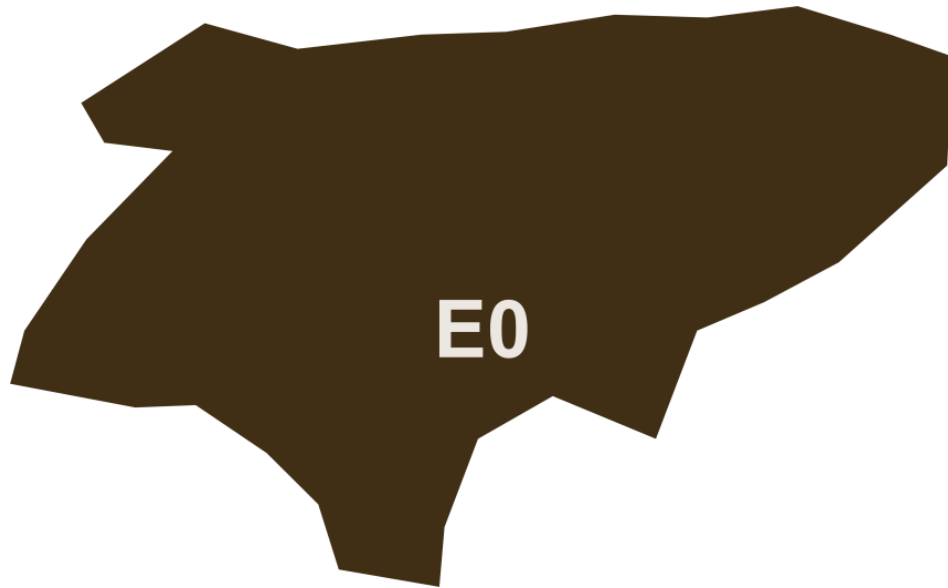
There are important fuel quality and environmental impact of vehicle emission challenges in the Region.

- The use of ethanol improves gasoline quality and creates flexibility in gasoline production.
- Ethanol use is a cost-effective way to increase gasoline octane and to replace more expensive gasoline components.
- Ethanol contributes to transport decarbonization and air quality improvement.
- There are opportunities across Latin America to increase the ethanol blend level and implement new policies on the use of gasoline-ethanol blends.

Sixteen countries with potential and additional use of ethanol were studied: 1) gasoline market profiles; 2) Optimization of gasoline blends with ethanol and 3) Environmental impact of gasolines blended with ethanol.



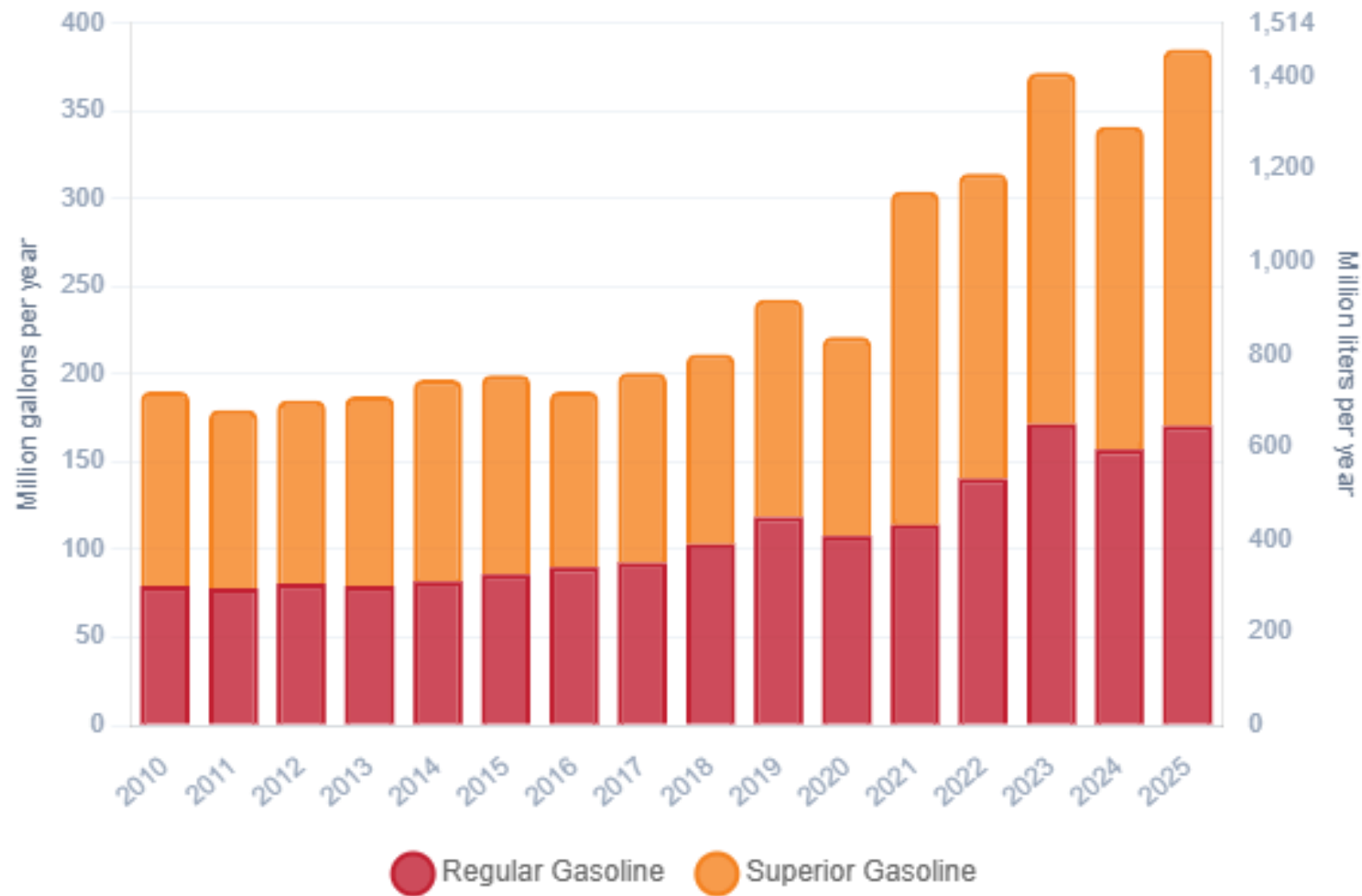
Ethanol Blending in Gasoline - Honduras



In 2025, gasoline demand was 385 million gallons (1,457 million liters). Regular gasoline RON 91 represented 44% while Superior gasoline RON 95 reached 56% of national demand. Honduras does not have gasoline production capacity and supplies its market with imports mainly from the United States.

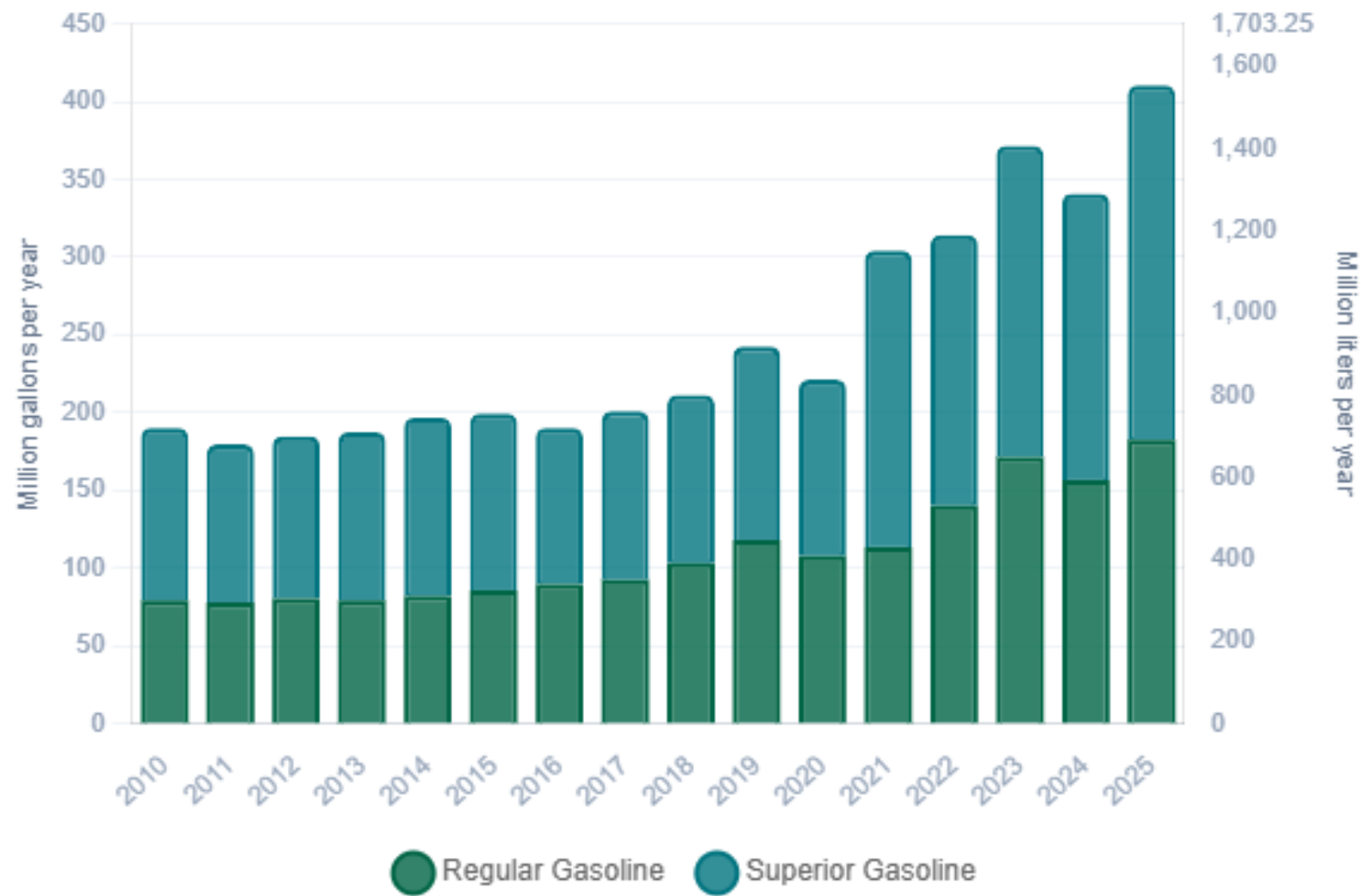
Honduras has no national mandate for ethanol blends. A new biofuel's law is under discussion.

Gasoline Demand in Honduras



Source: CEPAL, Secretaría de Energía Honduras

Gasoline Imports in Honduras



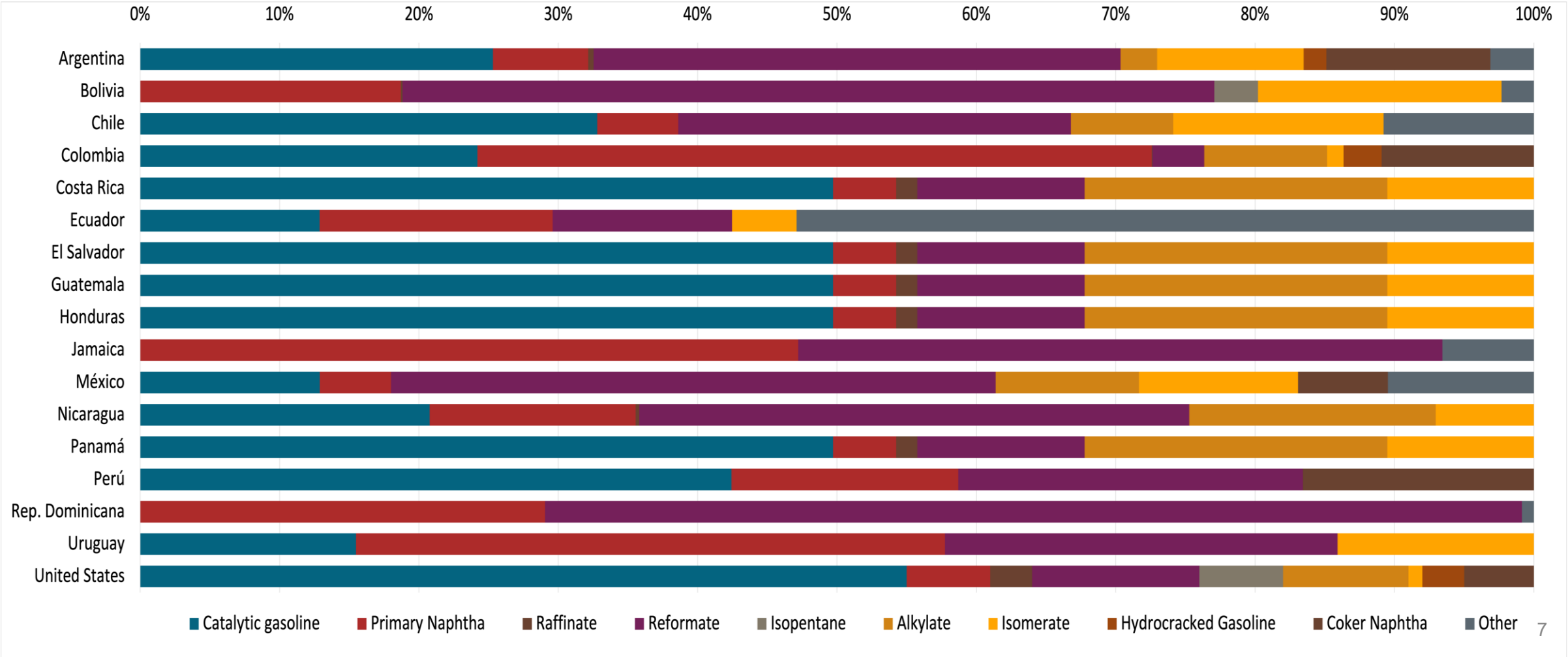
Source: CEPAL, Secretaría de Energía Honduras

Gasoline Quality in Honduras

Name	RTCA 75.01.08:19 p.m.	RTCA 75.01.19:19	EN 228:2012 + A1:2017 (Euro 6 enabling)			
Implementation Date	2021	2021	2017			
Applicability	Whole country	Whole country	All countries			
Selected Grade	Gasoline Premium	Gasoline Regular	RON 95 E5	RON 95 E10	RON 98 E5	RON 98 E10
Benzene Content	2,5% v/v	2,5% v/v	< 1 %v/v	< 1 %v/v	< 1 %v/v	< 1 %v/v
Aromatics	50% v/v	50% v/v	< 35 %v/v	< 35 %v/v	< 35 %v/v	< 35 %v/v
Olefins	30% v/v	30% v/v	< 18 %v/v	< 18 %v/v	< 18 %v/v	< 18 %v/v
Lead Content	< 0,013 g/l	< 0,013 g/l	< 5 mg/l	< 5 mg/l	< 5 mg/l	< 5 mg/l
Manganese	0,25% v/v	0,25% v/v	< 2,0 mg/l	< 2,0 mg/l	< 2,0 mg/l	< 2,0 mg/l
RON	> 95	> 91	> 95	> 95	> 98	> 98
MON	-	-	> 85	> 88	> 85	> 88
AKI						
Sulfur Content	< 500 mg/kg	< 500 mg/kg	< 10 mg/kg	< 10 mg/kg	< 10 mg/kg	< 10 mg/kg
Oxygen Content	2,7% v/v	2,7% v/v	<2,7 % m/m	<3,7 % m/m	<2,7 % m/m	<3,7 % m/m
Ethanol (EtOH)	-	-	<5 %v/v	<10 %v/v	<5 %v/v	<10 %v/v
RVP 37.8°C (Summer)	< 69 kPa	< 69 kPa	<> 60 - 70 kPa *Depends on the country, RVP is regulated in the EU Fuel Quality Directive			
RVP 37.8 °C(Winter)						
RVP 37.8°C (Transition)						
MTBE	-	-	-	-	-	-
Ethers 5 or more C Atoms	-	-	Based on oxygen content	<22 %v/v	Based on oxygen content	<22 %v/v

Gasoline Component Blending in Latin America

Gasoline is a blend of a base gasoline and other components. This blending is usually done at blending terminals as only 30% of the world's finished gasoline is distributed directly from refineries. Each component provides different properties to the final blend, for example, isomerates, alkylates and butanes increase the octane. The components commonly used in Latin America are:



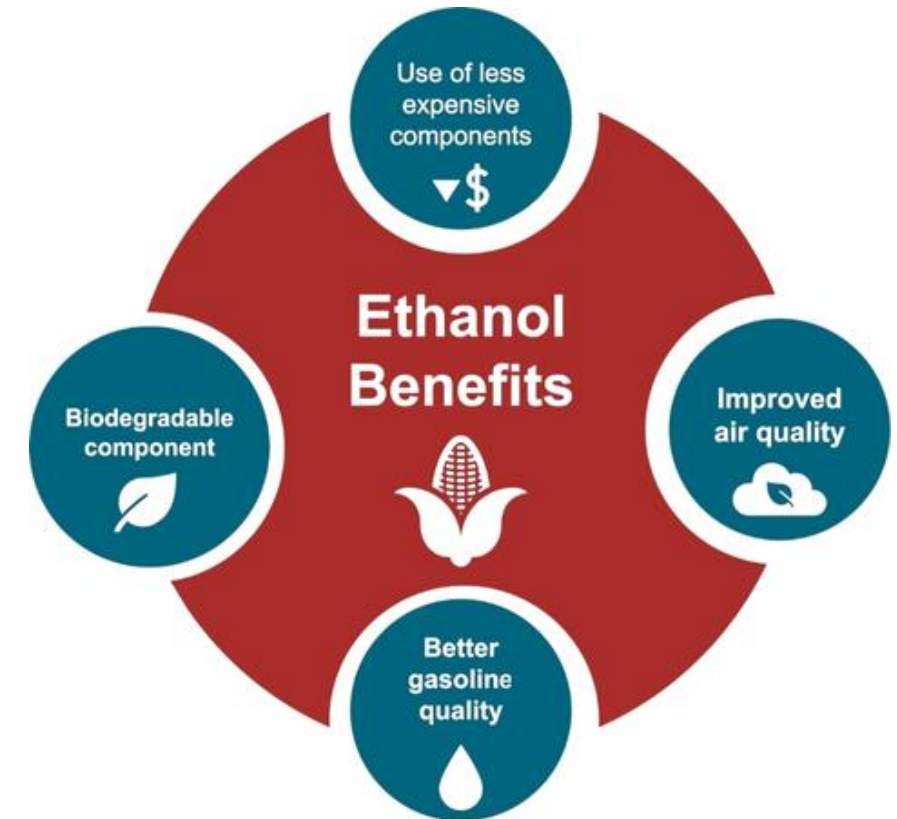
Gasoline Blending Optimization

In some parts of the world, ethanol is added to gasoline as a blending component. The advantages of ethanol include that it is a renewable fuel made of biomass; that it is an octane booster that helps to dilute sulfur; and that it allows the fulfillment of environmental objectives. To determine the optimal components to be blended with ethanol, a **blending model** was used. This model selects the components to add in the gasoline/ethanol blend based on:

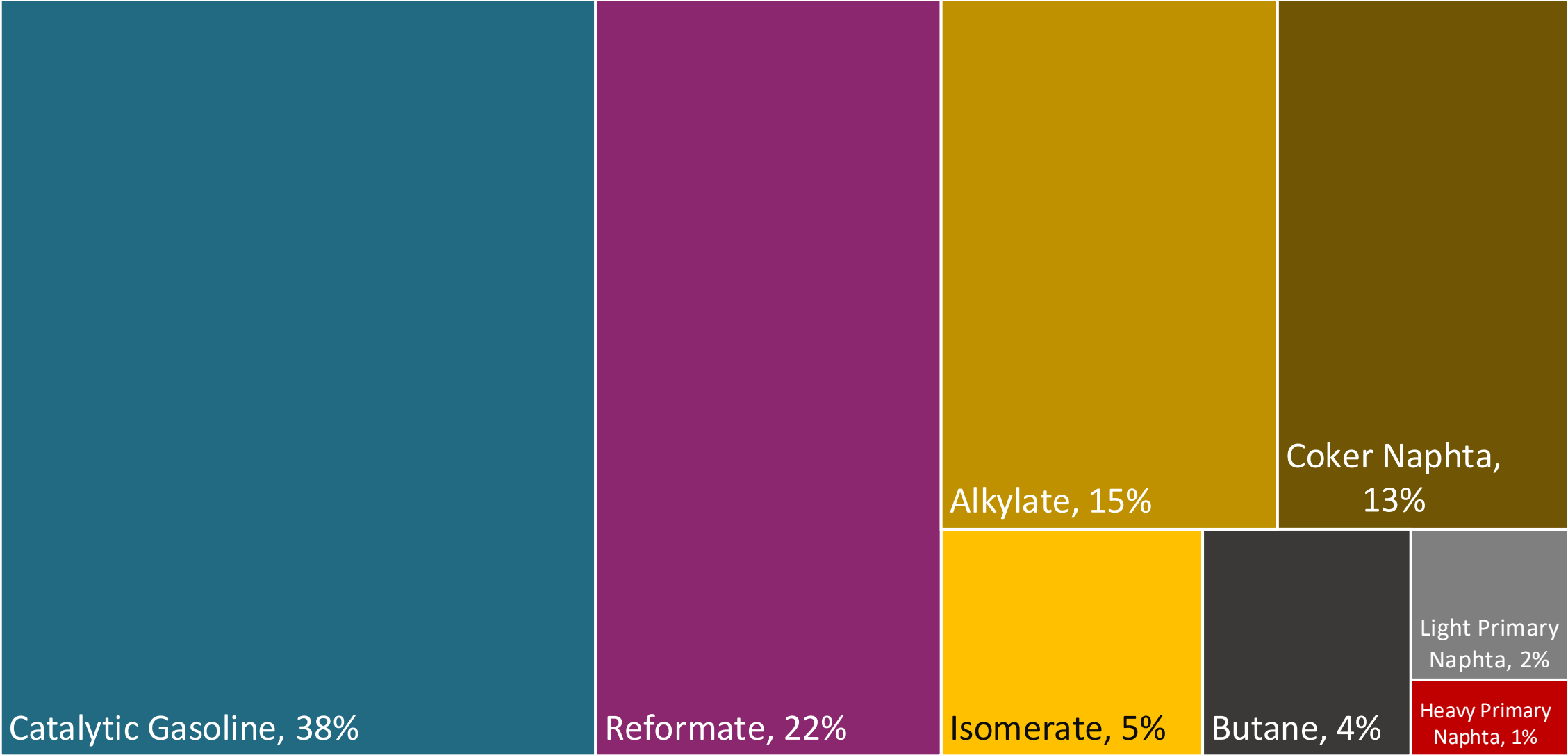
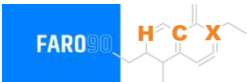
- Components prices,
- Properties each component affects,
- Quality parameters by country, and
- Component availability by country.

Through iterations, the model obtains the %v/v of the components to be blended with 10%, 15%, 20%, 25% and 30% of ethanol, in such a way that the final blend complies with the required properties of a finished gasoline by country.

The blending model uses gasoline component spot average prices 2025 and provides fuel prices that do not include country distribution costs, local taxes and subsidies and import or gas station margins.

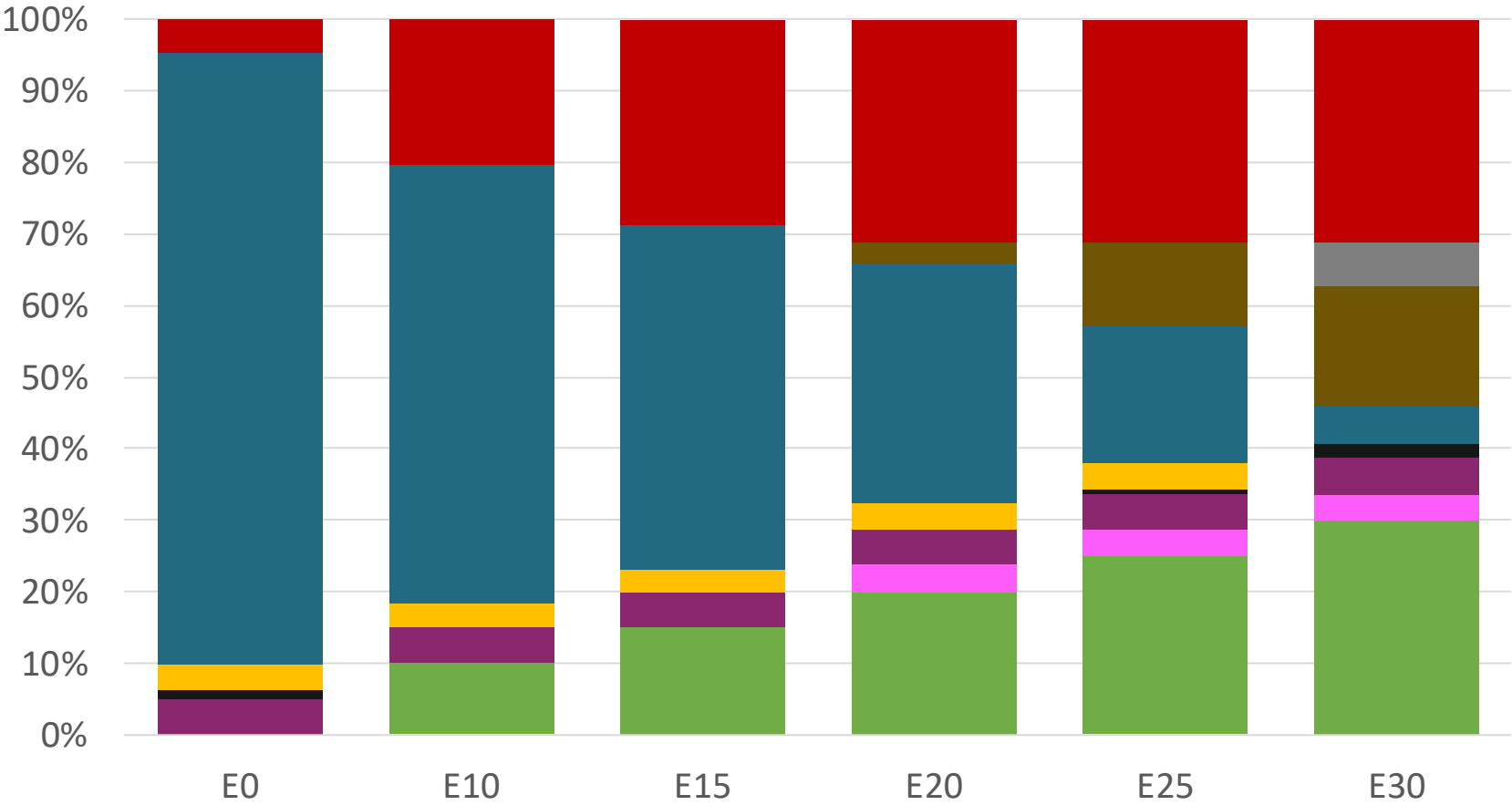
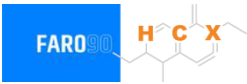


Blending Components - Honduras



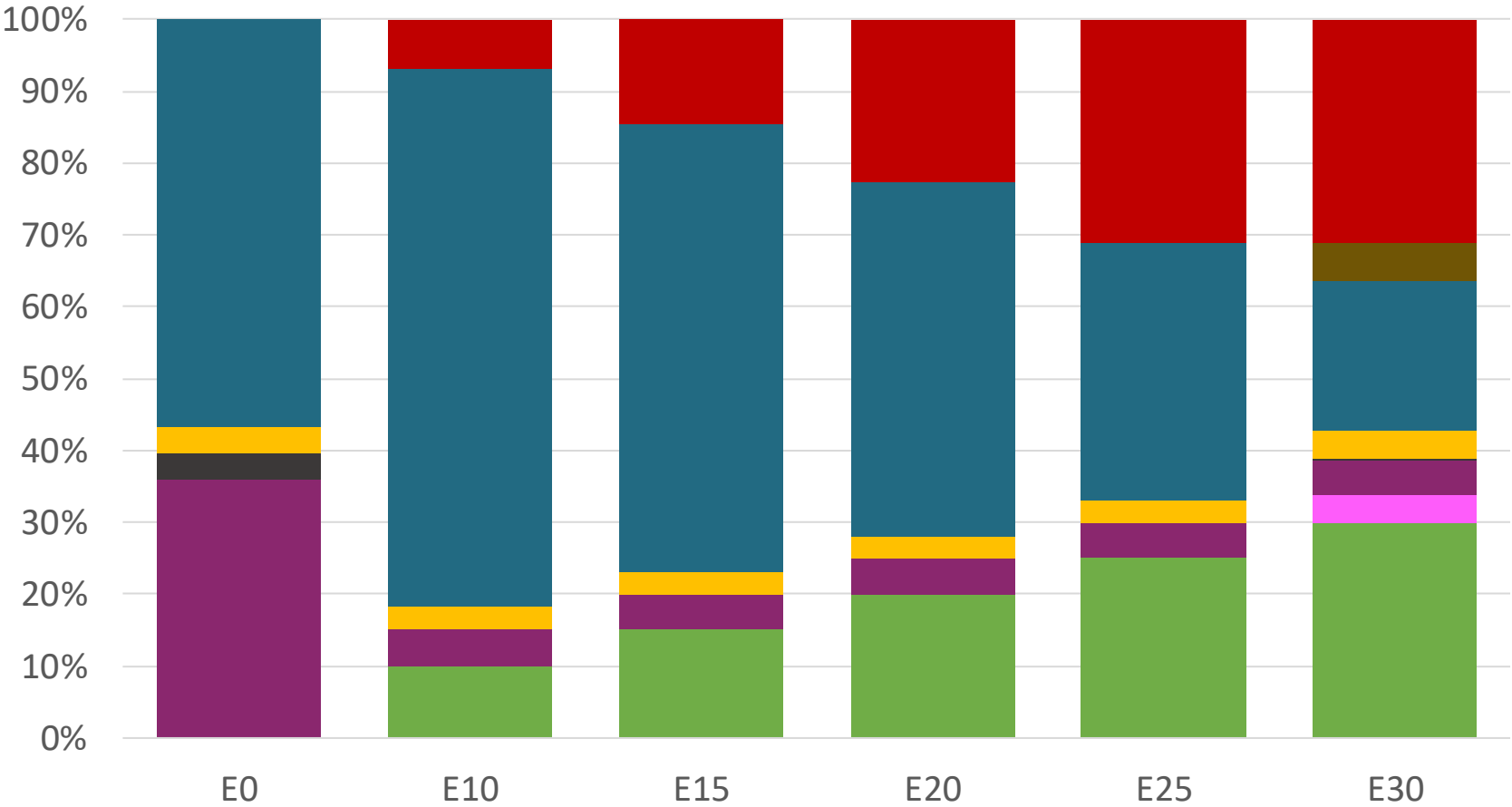
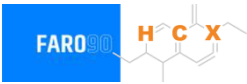
Source: Faro90

Honduras – Regular – Constant Octane



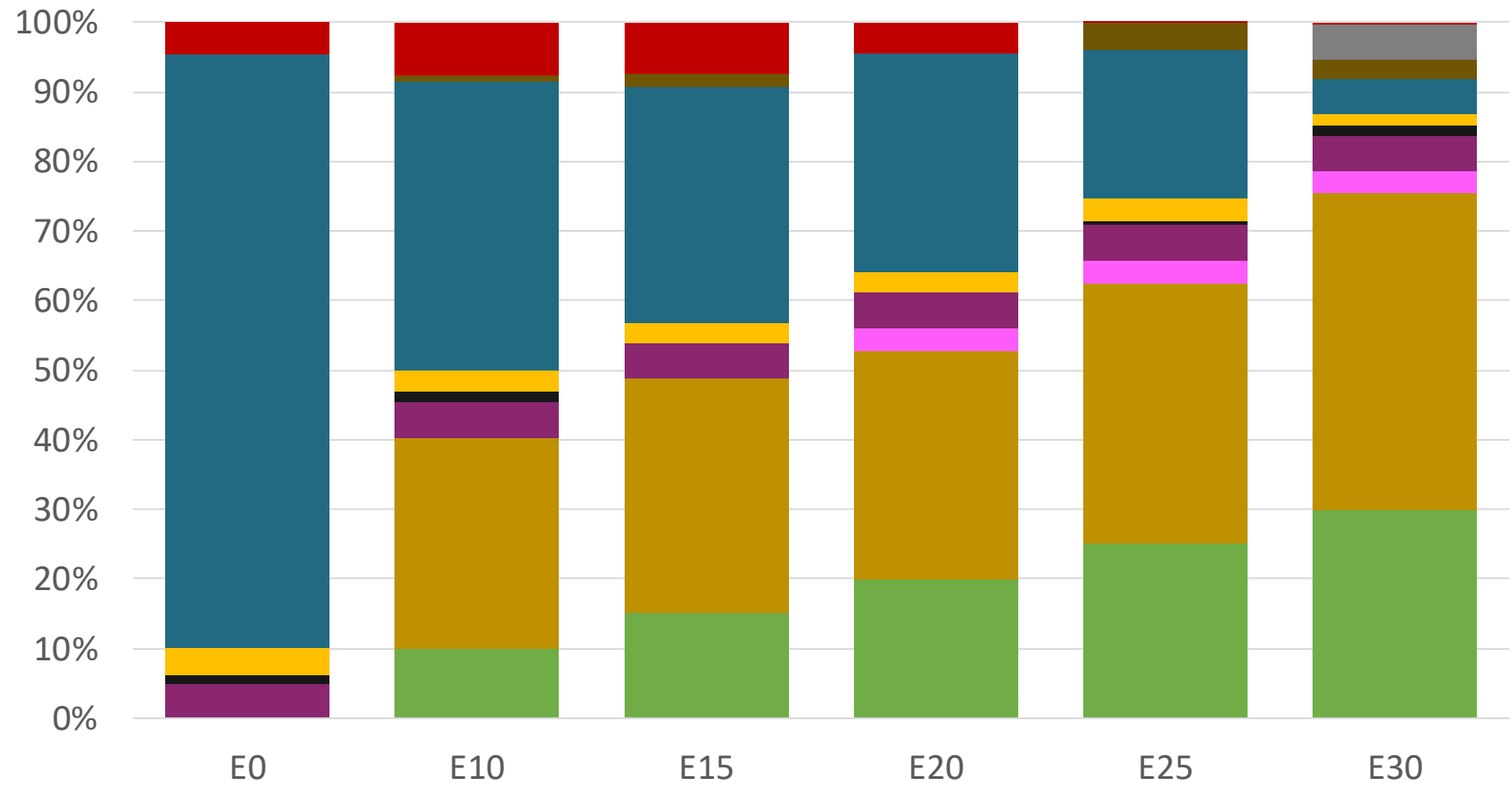
Octane (RON)	88.0	88.0	88.0	88.0	88.0	90.7
Price (USD/gal)	\$ 1.946	\$ 1.674	\$ 1.523	\$ 1.381	\$ 1.237	\$ 1.266

Honduras – Premium – Constant Octane



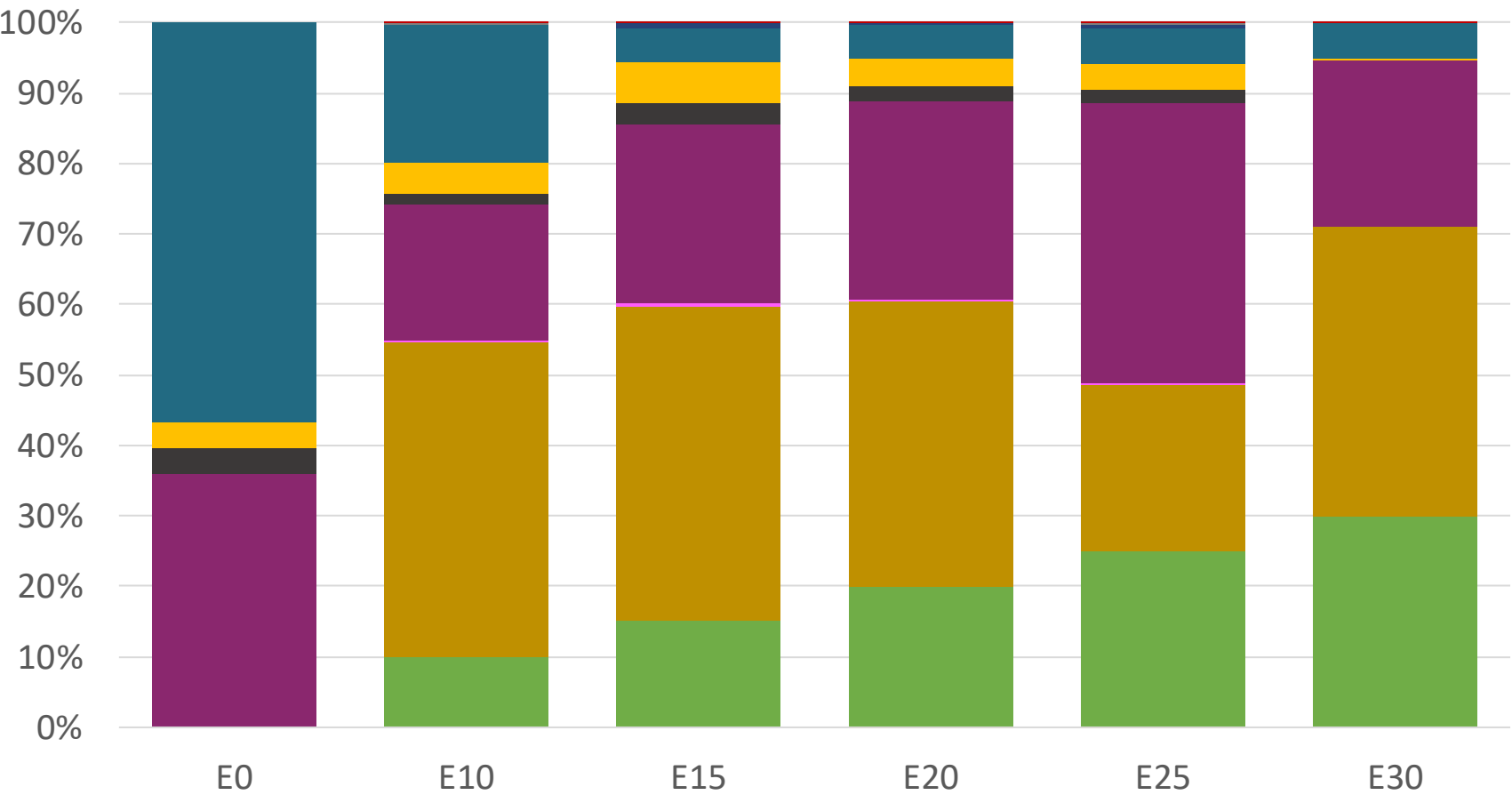
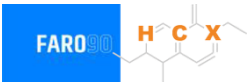
Octane (RON)	95.0	95.0	95.0	95.0	95.0	95.0
Price (USD/gal)	\$ 2.326	\$ 2.023	\$ 1.888	\$ 1.746	\$ 1.597	\$ 1.448

Honduras – Regular – Increased Octane



Octane (RON)	88.0	91.6	95.3	97.2	97.9	100.1
Price (USD/gal)	\$ 1.946	\$ 1.946	\$ 1.946	\$ 1.946	\$ 1.946	\$ 1.946

Honduras – Premium – Increased Octane



Octane (RON)	95.0	98.0	100.3	102.6	104.9	105.8
Price (USD/gal)	\$ 2.326	\$ 2.326	\$ 2.326	\$ 2.326	\$ 2.326	\$ 2.326

Vehicle Emission Impact for Ethanol Gasoline Blending

The model used in this analysis takes as a reference the [International Vehicle Emissions Model \(IVE\)](#).

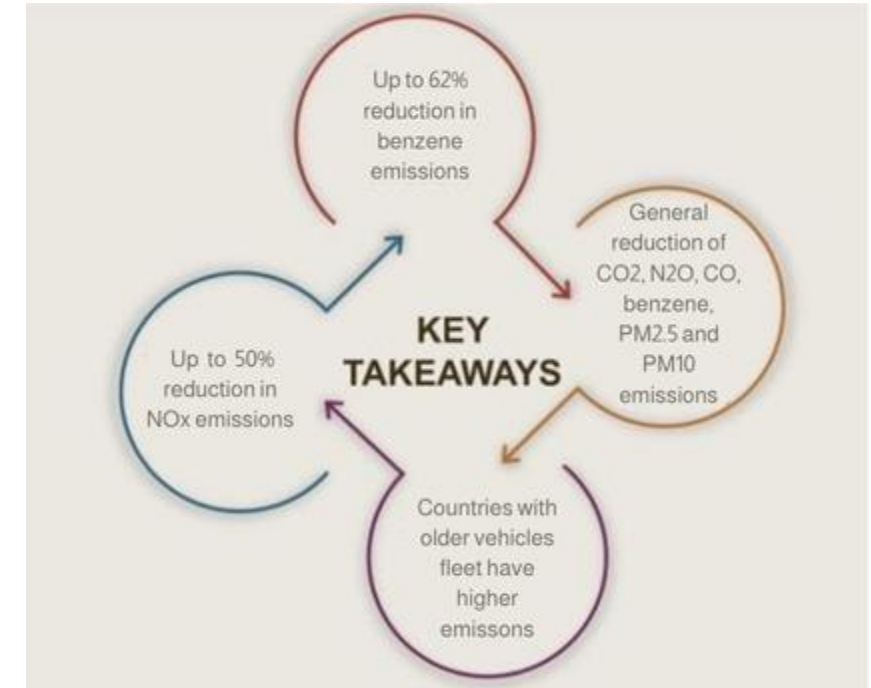
The model uses the Base Emission Rates from IVE model, as well as its Adjustment Factors based on:

- Vehicle technology (cars, trucks, buses, motorcycles),
- Vehicle fleet average age,
- Average traveled distance per vehicle by country, as well as
- Geographical and climatic conditions (altitude, humidity, temperature).

Emissions of criteria pollutants, toxic pollutants, and greenhouse gases (GHG) were calculated and calibrated with emission inventories, using real gasoline quality data. The reduction rates for gasoline/ethanol blends were obtained from various sources (IPCC, US Grains, among others).

Emission estimations for different pollutants for gasoline and gasoline/ethanol blends (10%, 15%, 20%, 25% and 30% ethanol) were determined using the IVE Model. A comparison between the results and the European (Euro 6) requirements is made. Results are also compared with real emissions of the United States vehicle fleet*.

Main Results



*Source: Bureau of transportation statistics.

Gasoline Vehicular Fleet in Honduras

Vehicle type	Engine	Exhaust	Recovery	Age	Number of vehicles						
Light automobiles	Carburator	3-Ways	PCV Tank	> 8 years, > 160 tkm	6,004						
	Single FI-Pt	3-Ways		> 8 years, > 160 tkm	20,322						
	Multi FI-Pt	3-Ways		> 8 years, > 160 tkm	147,410						
		Euro III		> 8 years, > 160 tkm	102,548						
		Euro IV		> 8 years, > 160 tkm	73,989						
		Euro V / Euro 6		> 8 years, > 160 tkm	55,318						
				4-8 years, 80-160 tkm	16,440						
				< 4 years, < 80 tkm	12,198						
	Carburator	3-Ways		PCV Tank	> 8 years, > 160 tkm	15,239					
	Single FI-Pt	3-Ways			> 8 years, > 160 tkm	138,486					
Medium automobiles	Multi FI-Pt	3-Ways	PCV Tank	> 8 years, > 160 tkm	309,468						
		Euro III		> 8 years, > 160 tkm	137,159						
		Euro IV		> 8 years, > 160 tkm	83,924						
		Euro V / Euro 6		> 8 years, > 160 tkm	82,046						
				4-8 years, 80-160 tkm	54,136						
				< 4 years, < 80 tkm	91,300						
		Small Engines		FI 4-cycles	None	PCV	> 8 years, > 160 tkm	23,510			
Euro II	> 8 years, > 160 tkm		29,000								
Euro III	> 8 years, > 160 tkm		298,892								
Euro IV	> 8 years, > 160 tkm		136,978								
Euro V	4-8 years, 80-160 tkm		181,607								
	4-8 years, 80-160 tkm		239,066								
	< 4 years, < 80 tkm		791,830								
Heavy	Carburator	3-Ways	PCV	> 8 years, > 160 tkm	7,503						
	FI	3-Ways		> 8 years, > 160 tkm	109,909						
		Euro III		> 8 years, > 160 tkm	36,890						
		Euro IV		> 8 years, > 160 tkm	18,371						
		Euro V / Euro 6		> 8 years, > 160 tkm	14,349						
				4-8 years, 80-160 tkm	13,125						
				< 4 years, < 80 tkm	29,889						

Vehicular Fleet: **3,046,872**

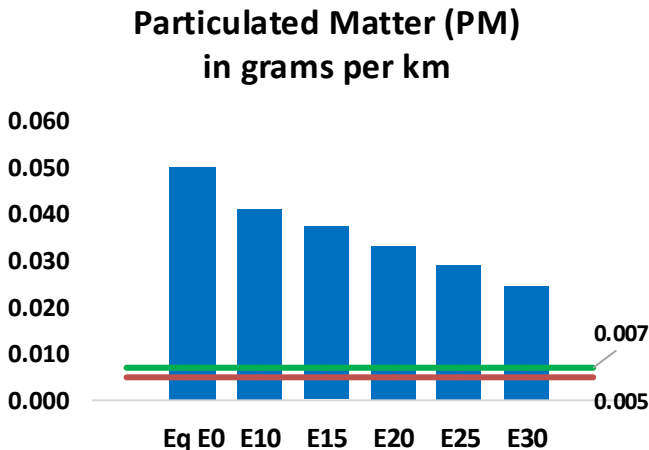
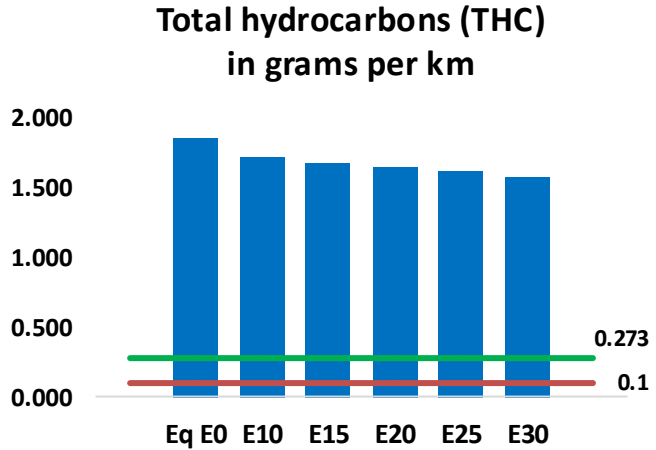
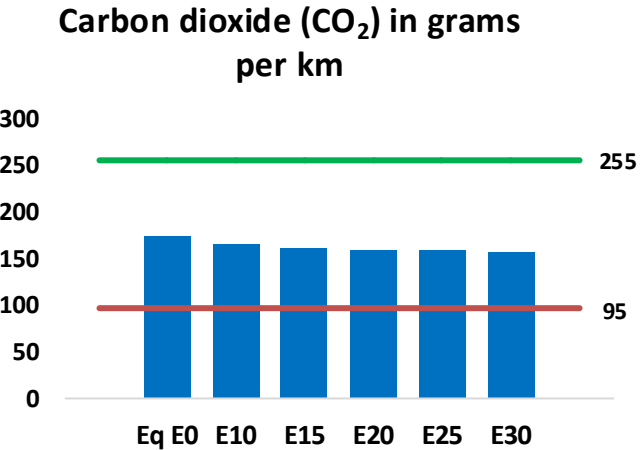
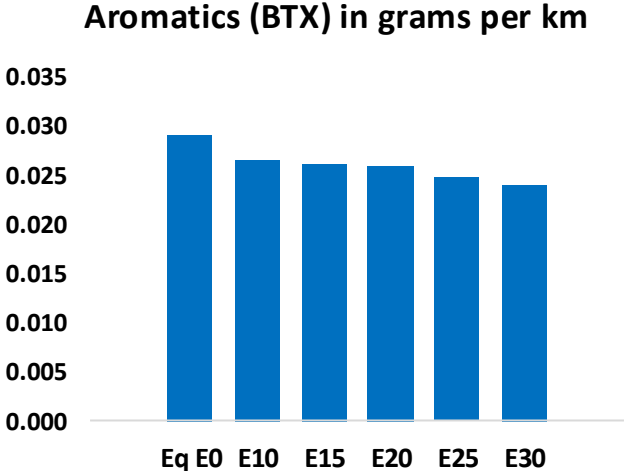
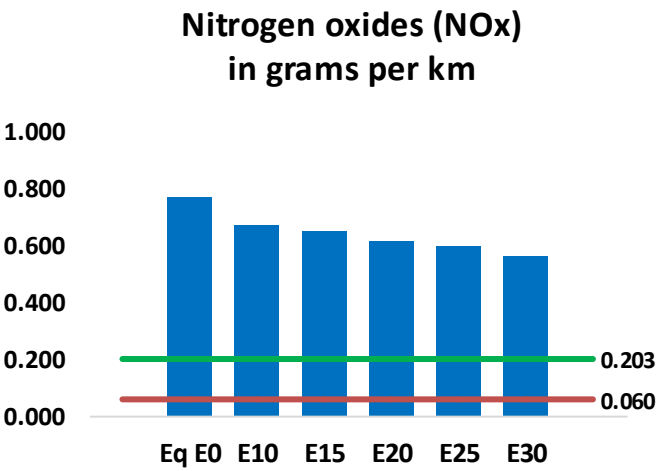
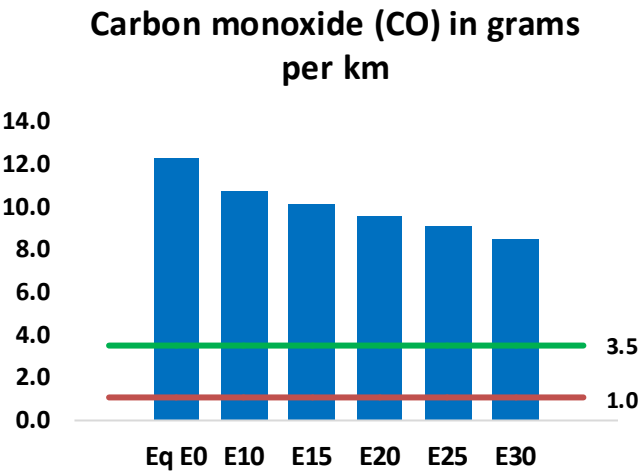
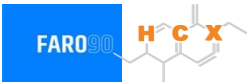
Average Age: **13 years**

Motorcycles: **50%**

Source: INE, analysis Faro 90

Honduras – Vehicular Emissions

United States
Euro 6



Honduras – Gasoline Vehicle Emissions

Vehicular Fleet Emissions (kg/day)	Eq E0	E5	E10	E15	E20	E25	E30	Reduction (%)	
								E10	E20
CO	657,884	616,144	574,405	541,926	510,034	485,751	455,533	-13%	-22%
CO2	9,243,575	9,005,883	8,781,396	8,604,844	8,517,479	8,496,255	8,339,644	-5%	-8%
VOC	98,921	95,343	91,764	89,617	87,818	86,544	84,459	-7%	-11%
NOx	41,297	38,131	36,011	34,958	33,232	31,893	30,308	-13%	-20%
SOx	108	100	92	85	78	71	64	-15%	-28%
NH3	3,656	3,620	3,584	3,601	3,641	3,670	3,693	-2%	0%
N2O	169	168	167	168	171	173	175	-1%	2%
CH4	22,145	22,145	22,145	22,477	22,964	23,385	23,784	0%	4%
Lead	-	-	-	-	-	-	-	0%	0%
Butadiene	366	343	319	301	283	270	251	-13%	-23%
Acetaldehyde	925	1,062	1,558	2,373	3,232	3,694	4,364	68%	249%
Formaldehyde	3,506	3,409	3,974	4,666	4,888	5,408	5,887	13%	39%
Benzene	1,559	1,498	1,420	1,397	1,386	1,325	1,282	-9%	-11%
PM2.5	551	501	451	407	364	317	269	-18%	-34%
PM10	2,132	1,938	1,745	1,574	1,407	1,228	1,041	-18%	-34%

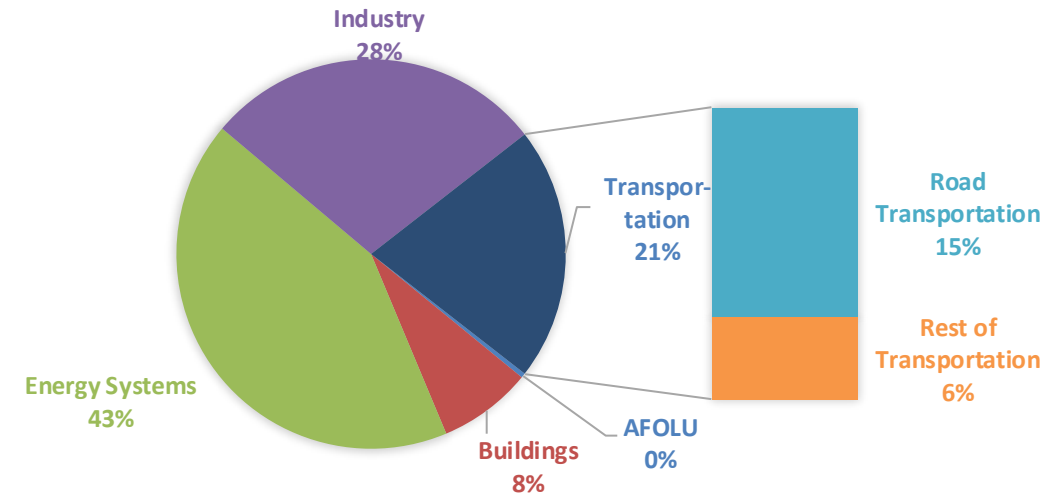
Life Cycle GHG Emissions

The Global Agenda for Climate Action calls for countries, cities and regions, businesses, and members of civil society around the world to achieve a low-carbon and climate-resilient economies in support of the Paris Agreement. Road Transportation accounts for 15% of total GHG emissions (IPCC,2023).

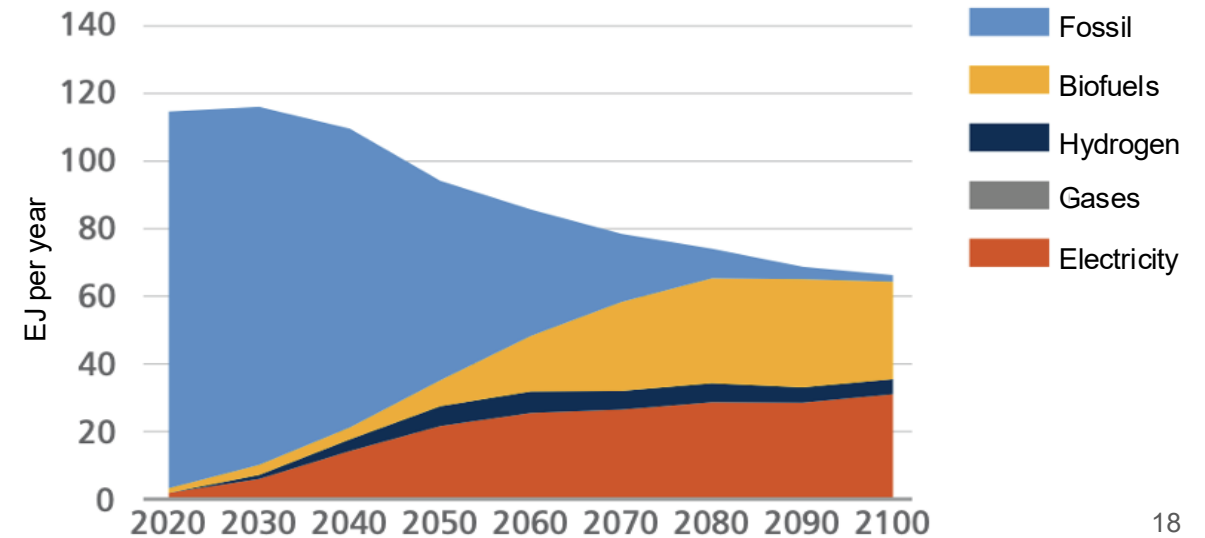
Ethanol gasoline blends are an internationally adopted practice that offer significant short-term benefits. Biofuels are an important solution to achieve reduction targets in the transportation sector.

Country GHG emission reductions are calculated using **RED II/III** and **GREET** life cycle emission methodologies and the **International Vehicular Emissions Model (IVE)**. Results are compared to the current country targets for 2035. Current 2025 country emission are those published by **IPCC**.

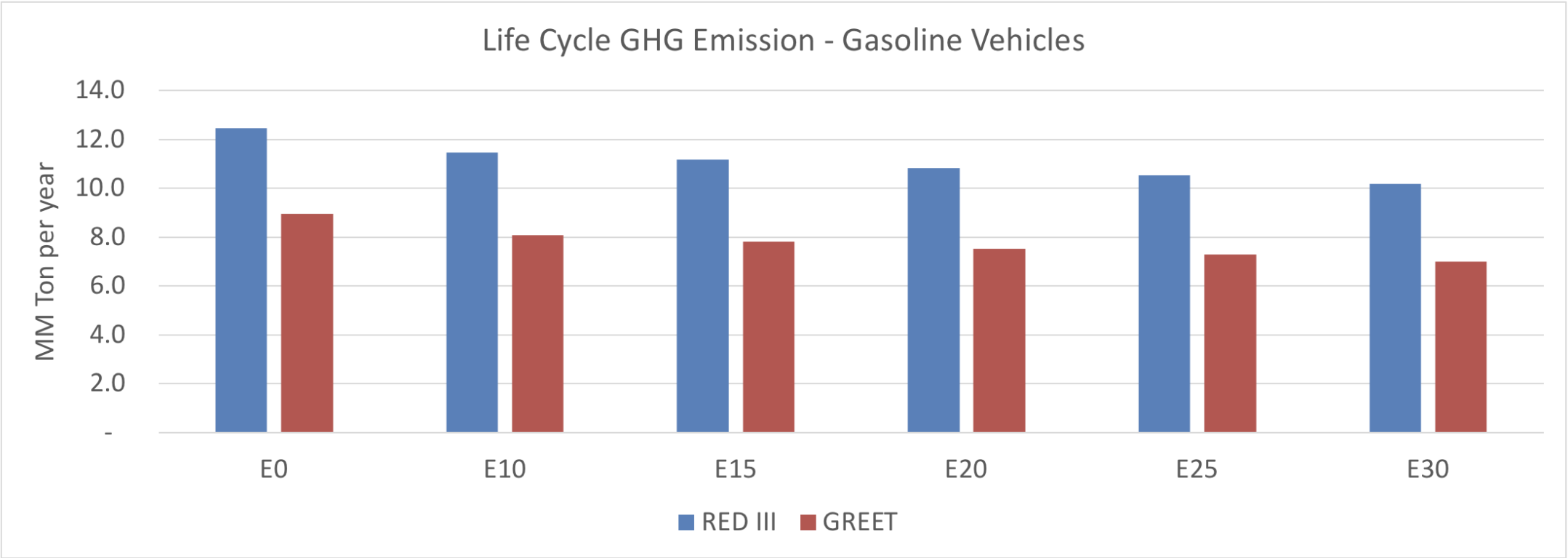
Current GHG Global Emissions Distribution



IPCC Transport Sector Sustainable Development Goals and climate policies Scenario, 2023



Honduras – Gasoline Vehicle Fleet GEI Emissions



Country Emissions 2025 (MMT/year)	30.7
2035 Target (MMT/year)	15.3
Delta	15.3

%Reduction vs E0	E0	E10	E15	E20	E25	E30
GREET 2025	0.0%	9.9%	12.7%	15.9%	18.5%	21.9%
RED II/III	0.0%	8.0%	10.4%	13.1%	15.3%	18.2%
% 2035 Target	E0	E10	E15	E20	E25	E30
GREET 2025	0.0%	5.8%	7.4%	9.3%	10.8%	12.8%
RED II/III	0.0%	6.5%	8.4%	10.7%	12.5%	14.7%